Janghyun **Noh**

AI Researcher

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Education

Myongji University
Seoul, S.Korea

M.S. in Artificial Intelligence — *Sep. 2024 - Aug. 2026*

- Conducted research at the Artificial Intelligence and Data Analytics Laboratory under the supervision of Prof. Minsik Oh.
- Conducted collaborative research with external institutions including Seoul National University Hospital (SNUH), the Korea Disease Control and Prevention Agency (KDCA), Kyungpook National University, and AIgenDrug Co., Ltd.
- Served as a teaching assistant for undergraduate courses including Deep Learning, Machine Learning, Data Mining, and Programming (Fall 2024 – Spring 2026).

Myongji University
Seoul, S.Korea

B.S. in Convergence Software and Data Technology — *Mar. 2018 - Aug. 2024*

- Conducted undergraduate research at the Artificial Intelligence and Data Analytics Laboratory under the supervision of Prof. Minsik Oh.
- Conducted collaborative research on clinical and genetic factors associated with hypertension progression in patients with hyperlipidemia in collaboration with the Korea Disease Control and Prevention Agency (KDCA).

RED School – Alternative Middle-High School
Geumsan, S.Korea

Student — *Mar. 2012 - Jan. 2018*

- Received academic excellence scholarships for two consecutive years.
- Participated in student leadership and community activities.

Research Experience

Artificial Intelligence and Data Analytics Lab, Myongji University
Seoul, S.Korea

Undergraduate Research Intern / Graduate Researcher — *Dec. 2022 - Aug. 2026*

- Developed scCap, an annotation-free phenotype prediction framework for single-cell RNA-seq data (first-author paper in Briefings in Bioinformatics), released with code and a Docker image.
- Built scRNA-seq models for MSI/MSS phenotype prediction and applied them to AI-based target discovery (with AIgenDrug).
- Conducted spatial transcriptomics analyses of TNBC Xenium datasets for cellular niche and tumor microenvironment characterization (with SNUH).
- Compared cirrhotic and non-cirrhotic HCC tissues using GeoMx spatial transcriptomics for immunomodulatory target discovery.
- Built a Biomni-based AI agent environment for analyzing CRISPR-Cas9-induced large DNA deletions.
- Developed multimodal COVID-19 severity prediction models combining MIMIC-IV clinical data and MIMIC-CXR chest X-rays (with KDCA).
- Implemented an nf-core/sarek variant-calling pipeline for apple whole-genome sequencing data (with Kyungpook National University).
- Designed measurement-based labels and prediction models for hypertension progression in KoGES cohort data (with KDCA).

Publications & Presentations

Journal Articles

Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data
Briefings in Bioinformatics (BIB)

Published — *2026*

J. Noh, Y. Shin, M. Kim, and M. Oh

Ferroptosis-based therapeutic strategy targeting iron metabolism and SLC7A11 overcomes oxaliplatin resistance in patient-derived pancreatic cancer organoids
Journal of Experimental & Clinical Cancer Research (JECCR)

Published — *2026*

J. Choi, K. Lee, M. Oh, **J. Noh**, Y. Kim, E. Hwang, I. Cho, W. Paik, J. Ryu, J. Ku, and S. Lee

Conference Presentations

Patient-Level Metacell Construction with Pre-trained Embeddings for Phenotype Prediction from scRNA-seq
KIISE Korea Computer Congress (KCC)

Oral Presentation — *2026*

Y. Shin, **J. Noh**, and M. Oh

Lightweight Analysis of Patient Phenotype Prediction Based on Single-Cell RNA-seq Data Reduction Techniques
KIISE Korea Software Congress (KSC)

Poster Presentation (Best Poster Award) — *2025*

Y. Shin, **J. Noh**, and M. Oh

An annotation-free framework for phenotype prediction using hierarchical multiple instance learning with single-cell RNA-sequencing data

International Conference on Genome Informatics ISCB-Asia 2025

Poster Presentation

J. Noh, Y. Shin, M. Kim, and M. Oh

Patient Phenotype Prediction from Single-cell RNA-seq Data through Annotation-free Multiple Instance Learning

BIOINFO 2025, Korean Society for Bioinformatics (KSBI) 2025

Poster Presentation (Excellent Poster Award)

J. Noh, Y. Shin, M. Kim, and M. Oh

AMIL: Automated Cell Grouping Framework for Phenotype Prediction with Multiple Instance Learning

KIISE Korea Computer Congress (KCC) 2025

Oral Presentation (Best Presentation Award)

J. Noh, M. Kim, and M. Oh

Predicting In-Hospital Mortality in Patients with Acute Respiratory Distress Syndrome Using MIMIC-III Data: A Comparative Study of Machine Learning Models with Varied Missing Data Handling Strategies

KIISE Korea Software Congress (KSC) 2023

Poster Presentation

J. Noh, G. Choi, and M. Oh

Honors & Awards

Awards

2026	**Research Honor Award**, Myongji University	*Seoul, S.Korea*
2025	**Best Poster Award**, KIISE Korea Software Congress (KSC)	*Seoul, S.Korea*
2025	**Excellent Poster Award**, BIOINFO 2025, Korean Society for Bioinformatics (KSBI)	*Seoul, S.Korea*
2025	**Best Presentation Award**, KIISE Korea Computer Congress (KCC)	*Seoul, S.Korea*
2022	**President's Merit Award**, Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*

Scholarships

2022	**TriBIG Project Scholarship**, Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*
2019	**Globalization Scholarship**, Myongji University	*Seoul, S.Korea*
2018	**Scholarship III**, Myongji University	*Seoul, S.Korea*
2016	**Academic Excellence Scholarship**, RED School	*Geumsan, S.Korea*
2015	**Academic Excellence Scholarship**, RED School	*Geumsan, S.Korea*

Projects

2024 – 2026	AI-based target discovery platform for CAR-Treg cell therapy in chronic inflammatory disease
2024 – 2026	Genetic and physiological mechanisms of apple blossom dormancy release under climate change
2024 – 2026	Spatial transcriptomics-based target discovery and validation for refractory solid tumors
2025 – 2026	Client-side web platform for detecting large CRISPR-Cas9-induced DNA deletions
2025 – 2026	GeoMx spatial transcriptomics of cirrhotic vs. non-cirrhotic HCC tissue
2025	COVID-19 severity prediction using multimodal clinical and imaging data
2023 – 2024	Hypertension-progression risk groups in hyperlipidemia patients (KoGES cohort)
2023	CatchYou: criminal composite sketch generation from witness descriptions
2022	Customs notice analysis and monitoring system for global supply-chain crises

Technical Skills

Programming	Python, R
Frameworks	PyTorch, TensorFlow, Scikit-learn, XGBoost, Scanpy, FastAPI
Tools	Linux, Git, Docker, Nextflow
Research	Machine Learning, Deep Learning, Multimodal Learning, Medical AI, Bioinformatics
Languages	Korean (Native), English (Full Professional Proficiency; OPIc IH 2026, TOEIC 920 2023)
Certifications	Engineer Information Processing (2023), SQL Developer – SQLD (2023)

Extracurricular Activities

Domestic

Google Developer Student Clubs, Myongji University

Seoul, S.Korea

AI Team Member

Oct. 2022 - Dec. 2023

- Engaged in machine learning and deep learning study groups
- Collaborated on AI projects and participated in AI competitions

Big Leader AI Academy

Seoul, S.Korea

Project Member

Jul. 2022 - Oct. 2022

- Developed a customs notice analysis and monitoring system in collaboration with the Big Data Team of KOTRA
- Participated in an industry-collaborative AI project for KOTRA internal service applications (TriBIG)
- Received a scholarship through outstanding performance in the TriBIG project

Data Youth Campus, Dongguk University

Seoul, S.Korea

Participant

Jul. 2022 - Aug. 2022

- Completed a data science-based intelligent software program affiliated with Korea Data Agency (KDATA)
- Conducted practical projects and training in machine learning and data analysis

International

Camino de Santiago (French Way)

Spain

Pilgrim

Sep. 2026 - Oct. 2026

Medical Mission Trip to the Philippines (GMMA)

Manila, Philippines

Volunteer

Jan. 2024

- Participated in medical volunteer activities with international healthcare professionals

International Work Camp in Iceland

Reykjavik, Iceland

Volunteer

Dec. 2019 - Jan. 2020

- Participated in environmental campaigns and intercultural exchange with international volunteers

Mission Trip to Vietnam

Hanoi, Vietnam

Volunteer

Aug. 2018

- Participated in intercultural exchange and community volunteer activities